① $dX_t = k(\underline{\theta - X_t})dt + \sigma dW_t$

$$\boxed{\begin{aligned} dX_t &= AX_t dt + B dW_t \\ X_t &= e^{At}X_0 + \int_0^t e^{A(t-s)} B\, dW_s \end{aligned}}$$

$$\boxed{du(t, X_t) = u'(t,X_t)dt + u_x(t,X_t)dX_t + \frac{1}{2}u_{xx}(t,X_t)\,dX_t dX_t}$$

$Y_t = \theta - X_t$

$dY_t = -dX_t = -k(\underbrace{\theta - X_t}_{Y_t})dt - \sigma dW_t = -kY_t dt - \sigma dW_t$

$u(t,x) = \theta - x$

$u' = 0$

$u_x = -1$

$u_{xx} = 0$

$Y_t = e^{-kt}Y_0 - \int_0^t e^{-k(t-s)}\sigma dW_s$

$\theta - X_t = e^{-kt}(\theta - X_0) - \int_0^t e^{-k(t-s)}\sigma dW_s$

$-X_t = -\theta + \theta e^{-kt} - X_0 e^{-kt} - \int_0^t \dots$

$X_t = X_0 e^{-kt} + \theta(1 - e^{kt}) + \int_0^t e^{-k(t-s)}\sigma dW_s$

② $dX_t = \mu X_t dt + \sigma X_t dW_t$

$df = f\cdot g$

$\frac{df}{f} = g$

$d\ln f = g$

$\ln X_t$

$Y_t = \ln X_t$

$dY_t = \frac{1}{X_t}dX_t - \frac{1}{2}\frac{1}{X_t^2}(dX_t)^2$

$dY_t = \frac{1}{X_t}(\mu X_t dt + \sigma X_t dW_t) - \frac{1}{2}\frac{1}{\cancel{X_t^2}}\sigma^2 \cancel{X_t^2}\, dt \ \textcircled{=}$

$$\begin{array}{c|cc} & dt & dW \\ \hline dt & 0 & 0 \\ dW & 0 & dt \end{array}$$

$(dX_t)^2 = \mu^2 X_t^2 (dt)^2 + 2\mu\sigma X_t^2 dt dW_t + \sigma^2 X_t^2 (dW_t)^2 = \sigma^2 X_t^2 (\underset{dt}{\underset{\parallel}{dW_t}})^2$

$\textcircled{=}\ \mu dt + \sigma dW_t - \frac{1}{2}\sigma^2 dt = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW_t$

$\rightarrow dY_t = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW_t \ \Big|\ \int_0^t \quad \Rightarrow Y_t = Y_0 + \int_0^t (\mu - \frac{\sigma^2}{2})ds + \int_0^t \sigma dW_s$

$Y_t - Y_0 = \int_0^t (\mu - \frac{1}{2}\sigma^2)ds + \int_0^t \sigma dW_s$

$Y_t - Y_0 = (\mu - \frac{1}{2}\sigma^2)t + \sigma(W_t - W_0)$

$X_t = \underbrace{X_0 e^{((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t)}}$

Метод интегрирующего множителя

$dX_t = \underbrace{f(t, X_t)}dt + \underline{\sigma(t)}\,\underline{X_t}\,dW_t$

① $dG_t = \sigma(t) G_t dW_t$

↘ где геом. броунс движ-е $\mu = 0$

$G_t = e^{-\frac{1}{2}\int_0^t \sigma^2(s)ds + \int_0^t \sigma(s)dW_s}$

② $F_t = \frac{1}{G_t}$

$dF_t = -\frac{1}{G_t^2}dG_t + \frac{2}{G_t^3}dG_t dG_t \ \textcircled{=}$

$f(t,x) = \frac{1}{x}$

$f' = 0$

$f_x = -\frac{1}{x^2} \qquad f_{xx} = \frac{2}{x^3}$

$\textcircled{=}\ -\frac{1}{G_t^{\cancel{2}}}\sigma(t)\cancel{G_t}dW_t + \frac{1}{\cancel{2}}\frac{\cancel{2}}{G^{\cancel{3}}}\sigma^2(t)\cancel{G_t^2}dt =$

$= -\frac{1}{G_t}\sigma(t)dW_t + \frac{1}{G_t}\sigma^2(t)dt =$

$= F_t\sigma^2(t)dt - \sigma(t)F_t dW_t$

③ $C_t = F_t X_t$

$dC_t = d(F_t X_t) = F_t dX_t + X_t dF_t + dX_t dF_t =$

$= F_t(\underline{f(t,X_t)dt + \sigma(t)X_t dW_t}) + \underline{X_t(F_t\sigma^2(t)dt - \sigma(t)F_t dW_t)}$

$-\sigma^2(t)X_t F_t dt =$

$= \underline{F_t f(t,X_t)dt} + \cancel{\sigma(t)F_t X_t dW_t} + \cancel{X_t F_t \sigma^2(t)dt} - \cancel{\sigma(t)X_t F_t dW_t} - \cancel{\sigma^2(t)X_t F_t dt}$

$d(C_t) = F_t f(t, \underbrace{X_t})dt$

$\frac{\underline{X_t \cdot G_t} \cdot F_t}{C_t\ G_t}$

$\underline{dC_t = F_t f(t, C_t G_t)dt}$ "детерм." ОДУ

④ $\underline{X_t = G_t C_t}$

Пример $dX_t = \frac{1}{X_t}dt + \underline{\alpha X_t dW_t}$

$f(t,x) = \frac{1}{x}$

$G_t = e^{-\frac{\alpha^2}{2}t + \alpha W_t}$

$F_t = \frac{1}{G_t} = e^{\frac{\alpha^2}{2}t - \alpha W_t}$

$dC_t = F_t f(t, C_t G_t) = F_t\cdot\frac{1}{C_t G_t} = \frac{F_t^2}{C_t}$

$\boxed{dC_t = \frac{F_t^2}{C_t}}$

$\underbrace{C_t dC_t} = F_t^2$

$\frac{1}{2}dC_t^2 = F_t^2$

$C_t^2 = \int_0^t F_s^2 ds + \_\_ = 2\int_0^t e^{\alpha^2 s - 2\alpha W_s}ds + \_\_$

$X_t = G_t C_t = e^{\alpha W_t - \frac{\alpha^2}{2}t}\sqrt{2\int_0^t e^{\alpha^2 s - 2\alpha W_s}ds + \_\_}$

$\boxed{4}$ $dX_t = k(\theta - X_t)dt + \sigma\underline{\sqrt{X_t}}dW_t$

$E[X_t] = ?$

$X_t = X_0 + \int_0^t k(\theta - X_s)ds + \int_0^t \sigma\sqrt{X_s}dW_s$

$\underline{E[X_t]} = E[X_0] + kE[\int_0^t(\theta - X_s)ds] + \cancel{E[\int_0^t \sigma\sqrt{X_s}dW_s]} \rightarrow 0$

$= E[X_0] + k(\theta t - \int_0^t E[X_s]ds)$

$Y_t = X_0 + k(\theta t - \int_0^t Y_s ds)$

$\underline{dY_t = k\theta - kY_t} \rightarrow$ ОДУ $Y \qquad Y_0 = X_0$

$Y_t = \theta + (X_0 - \theta)e^{-kt}$

$E(X_t) = \theta + (X_0 - \theta)e^{-kt}$

$E[X^2]$

$dX^2 = 2XdX + (dX)^2 \rightarrow$ интегр $\rightarrow E \rightarrow$ ОДУ $\rightarrow E[x^2]$

$X_{t_{i+1}} = X_{t_i} + \underbrace{\int_{t_i}^{t_{i+1}} \mu X_s ds} + \int_{t_i}^{t_{i+1}} \sigma X_s dW_s \qquad X_0 = x$

$X_{t_i} + \mu X_{t_i}(t_{i+1} - t_i) + \sigma X_{t_i}(W_{t_{i+1}} - W_{t_i})$

$\downarrow$

$\sim \underbrace{N(0, t_{i+1} - t_i)}$ V

$X_t = X_0 e^{((\mu - \frac{1}{2}\sigma^2)t + \sigma \underbrace{W_t})}$

$W_t = \lim \sum_{i=1}^N \xi_i \qquad \xi_i \sim N(0, \Delta t)$

$(0, T) \rightarrow N \qquad \Delta t = \frac{T}{N}$

Сгенерир. N точек из $\sim N(0, \Delta t)$

cumsum of точек $W_t \rightarrow$ одна реализация X_t

N точек для аналит. решения Δt

$N/64$ точек $\qquad \Delta t \times 64$

$N/16$ точек $\Big\}$ для Э-М. $\qquad \Delta t \times 16$

$N/2$ точек $\qquad \Delta t \times 2$